S VENKATADRI

Software Engineer | AI/ML Engineer | Backend Engineer
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PROFESSIONAL SUMMARY

Software Engineer and AI/ML Engineer with **4+ years of experience** designing, developing, and deploying scalable cloud-native applications, backend systems, and AI-powered enterprise solutions for large-scale business environments. Proven expertise in building high-performance microservices, REST APIs, and distributed systems while delivering secure, production-ready software that improves application performance, operational efficiency, and business outcomes. Experienced in applying **Machine Learning, Generative AI, Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG)** to develop intelligent automation and enterprise AI solutions. Strong background in software architecture, system design, cloud technologies, CI/CD, and the Software Development Life Cycle (SDLC), with a focus on writing clean, maintainable, and scalable code. Collaborative engineering professional passionate about solving complex technical challenges, optimizing software performance, and delivering innovative solutions through Agile development and cross-functional teamwork.

TECHNICAL SKILLS

Programming Languages: Python, C#, JavaScript, TypeScript, SQL
AI & Machine Learning: Machine Learning, Deep Learning, Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Scikit-learn, PyTorch, Azure OpenAI, Semantic Search, Embeddings, Vector Databases, Model Evaluation
Backend Development: FastAPI, Flask, REST APIs, Microservices, Distributed Systems, Object-Oriented Programming (OOP), Design Patterns, Asynchronous Programming, Authentication, Authorization, Role-Based Access Control (RBAC).
Cloud & DevOps: Microsoft Azure, Azure DevOps, Docker, Kubernetes (AKS), AWS Lambda, Amazon S3, Jenkins, Git, GitHub, CI/CD
Databases: SQL Server, PostgreSQL, Azure SQL Database, NoSQL(MongoDB, DynamoDB)
Data Engineering: Apache Kafka, Azure Data Factory, Azure Synapse Analytics, ETL Pipelines, Data Integration
Software Engineering: Software Development Life Cycle (SDLC), Agile, Scrum, System Design, Software Architecture, Performance Optimization, Debugging, Unit Testing, Code Reviews, Integration Testing, Regression Testing
Monitoring & Observability: Azure Monitor, Application Insights, Logging, Troubleshooting, Root Cause Analysis
Frontend: React.js, TypeScript, Tailwind CSS

PROFESSIONAL EXPERIENCE

Deloitte — **Feb 2025 - Present**
AI/ML Engineer — *Kansas City, KS, USA*

- Architected and developed AI-powered enterprise applications supporting **2,000+ users** and processing **500,000+ business documents**, enabling intelligent document processing, semantic search, automated knowledge retrieval, and workflow automation across enterprise repositories.
- Engineered scalable **FastAPI** microservices and secure REST APIs for document ingestion, metadata extraction, intelligent indexing, and AI-driven workflows, reducing document processing latency by **35%** while improving scalability and maintainability.
- Designed and implemented **Retrieval-Augmented Generation (RAG)** solutions using **Azure OpenAI, LangChain, embeddings, and vector databases**, improving contextual response accuracy by **40%** and significantly reducing manual research efforts.
- Developed and evaluated Machine Learning and Generative AI solutions using **Scikit-learn, PyTorch, Prompt Engineering, and LLM evaluation techniques** to enhance document classification, response quality, and intelligent automation while improving user satisfaction.
- Integrated enterprise applications with **Azure OpenAI, AWS Lambda, Amazon S3, and secure REST APIs**, enabling scalable AI-powered document intelligence, serverless processing, and secure cloud-based content retrieval across enterprise systems.
- Built high-throughput backend processing pipelines utilizing **Apache Kafka**, asynchronous programming, and distributed microservices, improving event-driven document processing, increasing system throughput by **40%**, and ensuring reliable communication between enterprise services.
- Optimized backend application performance through efficient API design, caching strategies, vector indexing, SQL optimization, and distributed processing, reducing average API response times by **40%** while improving application reliability and scalability.
- Containerized cloud-native applications using **Docker** and **Kubernetes (AKS)** while automating build, testing, and deployment pipelines through **Azure DevOps** and **Jenkins**, improving deployment consistency, accelerating release cycles, and enhancing software quality.

- Partnered with architects, product managers, data engineers, and cross-functional stakeholders throughout the **Software Development Life Cycle (SDLC)** to deliver scalable AI-powered enterprise solutions, produce technical documentation, troubleshoot production issues, and continuously optimize application performance. analysis.

Quadrant Technologies **Nov 2021 - July 2024**
Software Engineer | Cloud & Backend Solutions Engineer *Hyderabad, India*

- Developed and maintained cloud-native enterprise applications using **Python, Microsoft Azure, SQL Server, and PostgreSQL**, supporting large-scale analytics platforms and improving application reliability for business-critical operations.
- Designed, developed, and deployed **30+ RESTful APIs** and backend microservices using **FastAPI, Flask, Python, and SQL**, enabling seamless integration between enterprise applications, third-party services, and cloud platforms.
- Automated business workflows using **Python, Azure Functions, and cloud automation services**, reducing manual effort by **30%**, improving operational efficiency, and minimizing processing errors.
- Engineered scalable **ETL pipelines** using **Azure Data Factory, Azure Synapse Analytics, SQL Server, and Apache Kafka**, enabling reliable data integration and high-volume enterprise reporting.
- Optimized SQL queries, stored procedures, indexing strategies, and database operations across **SQL Server, PostgreSQL, and MongoDB**, improving application performance by **35%** and reducing data retrieval latency.
- Containerized backend applications using **Docker** and **Kubernetes (AKS)** while supporting deployment automation through **Azure DevOps, Jenkins, and CI/CD pipelines**, improving deployment consistency and release efficiency.
- Implemented centralized monitoring, logging, and observability using **Azure Monitor** and **Application Insights**, reducing incident resolution time by **40%** through proactive monitoring and root cause analysis
- Collaborated with architects, business analysts, and cross-functional engineering teams throughout the **Software Development Life Cycle (SDLC)**, contributing to system design, testing, production support, code reviews, and AI-driven proof-of-concept initiatives using **Scikit-learn** and **PyTorch**. productivity.

PROJECTS

AI-Powered Cloud Document Intelligence Platform
Python | FastAPI | Azure OpenAI | LangChain | RAG | Docker | Kubernetes

- Architected and developed an enterprise AI-powered document intelligence platform capable of processing and indexing **500,000+ documents** using **FastAPI, Azure OpenAI, LangChain, and cloud-native microservices**, improving enterprise knowledge discovery and operational efficiency.
- Implemented a **Retrieval-Augmented Generation (RAG)** architecture using **LLMs, embeddings, vector databases, semantic search, Scikit-learn, and PyTorch**, improving contextual response accuracy by **40%** and reducing manual document search efforts.
- Automated secure cloud deployment using **Docker, Kubernetes, Azure DevOps, Jenkins, AWS Lambda, and Amazon S3**, implementing authentication, RBAC, logging, monitoring, and CI/CD pipelines for scalable production deployments.

Library Management System
React.js | Node.js | MongoDB | REST APIs

- Developed a full-stack Library Management System using **React.js, Node.js, MongoDB, and REST APIs**, supporting inventory management, reservations, borrowing workflows, user administration, and real-time resource tracking.
- Designed secure backend services with **authentication, authorization, role-based access control (RBAC)**, and optimized MongoDB queries, improving query performance by **35%** while ensuring secure transaction processing.
- Built automated notification workflows, overdue tracking, reporting dashboards, and inventory monitoring capabilities, reducing manual administrative effort and enhancing the overall user experience.

EDUCATION

University of Central Missouri | Warrensburg MO
Master's, Computer Science

CERTIFICATIONS

- Microsoft Azure Administrator AZ-104
- Microsoft Azure Fundamentals AZ-900
- Google AI Essentials
- IBM Generative AI Essentials
- DeepLearning.AI Prompt Engineering for Developers
- Machine Learning using Python